

CURRICULUM VITAE

09/01/2026

Yujie Zheng

Department of Earth Sustainable Earth Systems Sciences

School of Natural Science and Mathematics

University of Texas at Dallas

Educational History:

Ph.D., 2020, Stanford University, Stanford, CA, USA, Geophysics

Bachelor of Science, 2014, Peking University, Beijing, China, Geophysics

Employment History

Assistant Professor, 2023 - present, University of Texas at Dallas, Richardson, TX, USA

Postdoctoral Scholar, 2020 – 2023, California Institute of Technology, Pasadena, CA, USA

Graduate Research Assistant, 2014 – 2020, Stanford University, Stanford, CA, USA

Professional recognitions and honors (study, teaching, research, service):

Outstanding Student Paper Award, 2017, American Geophysics Union

The Joshua L. Soske Fellowship, 2014, Stanford University

Principle's Award for undergraduate research, 2012, Peking University

Chinese Universities Study Award, 2012, National University of Singapore

The May 4th Fellowship, 2010, Peking University

Professional memberships:

Member, American Geophysical Union, 2014 – present

Member, Institute of Electrical and Electronics Engineers, 2025 -- present

Achievements in original investigation:

Articles in refereed journals:

[13] Jones, C.E., Rosenqvist, A., Rommen, B., Fitzyk, M., Rignot, E., Scheuchl, B., **Zheng, Y.**, Hooper, A., Simons, M., Kobayashi, T., Siqueira, P., Santoro, M., Tadono, T., Hess, L., McNairn, H., Frulla, L., Johansson, M., Dierking, W., Rouyet, L., Bartsch, A., Lopez Dekker, P., Johannessen, J., (2026), Observation and coordination needs for current, near-future, and next generation Earth-observing SAR systems, *Earth and Space Science*, accepted.

[12] **Zheng, Y.**, Fattahi, H., (2026) Modeling, Prediction and Retrieval of Surface Soil Moisture from InSAR Closure Phase, *Remote Sensing of Environment*, 333, 115104.

<https://doi.org/10.1016/j.rse.2025.115104>

[11] **Zheng, Y.**, Simons, M, (2025), Integrating long-term InSAR monitoring into local groundwater management: Insights from the Hollywood Basin, California, *Water Resources Research*, 61(11), e2024WR039161. <https://doi.org/10.1029/2024WR039161>

[10] Tay, C., Jovani-Sancho, A., Yulianti, L., Evans, C., Callaghan, N., Jaya A., Salman, R., **Zheng, Y.**, Susilo, S., Dohong, S., Yun, S. (2025), Satellite radar advances carbon emissions accountability over tropical peat, *Communications Earth & Environment*, 6(1), 971. <https://doi.org/10.1038/s43247-025-02926-6>

[9] Zhang, S., Luo, B., **Zheng, Y.**, Lumley D., Sen, M., Zhu, H., (2025), Mapping spatiotemporal variations of near-surface seismic velocity to monitor groundwater in central Oklahoma, *Geophysical Research Letters*, <https://doi.org/10.1029/2025GL115201>

- [8] Mao, S., Ellsworth, W., **Zheng, Y.**, Beroza, G., (2025), Depth-Dependent Seismic Sensing of Groundwater Recovery from the Atmospheric-River Storms of 2023, *Science*, <https://doi.org/adr6139>
- [7] **Zheng, Y.**, Fattahi, H., Agram, P., Simons, M., and Rosen, P., (2022), On closure phase and Systematic Bias in Multi-looked SAR Interferometry. *IEEE Transactions on Geoscience and Remote Sensing*, <http://doi.org/10.1109/TGRS.2022.3167648>
- [6] **Zheng, Y.**, Laura Blackstone and Segall, P., (2022), Constraints on absolute magma chamber volume from geodetic measurements of Trapdoor faulting in the Galapagos. *Geophysical Research Letters*, <https://doi.org/10.1029/2021GL095683>
- [5] Wang, T., **Zheng, Y.**, Pulvirenti, F., Segall, P., (2021). Post-2018 caldera collapse reinflation uniquely constrain Kilauea's magmatic system, *Journal of Geophysical Research: Solid Earth*, <https://doi.org/10.1029/2021JB021803>
- [4] **Zheng, Y.**, Zebker, H.A., and Michaelides, R.J., (2021). A New Decorrelation Phase Covariance Model for Noise Reduction in Unwrapped Interferometric Phase Stacks, *IEEE Transactions on Geoscience and Remote Sensing*, <https://doi.org/10.1109/TGRS.2021.3050087>
- [3] Michaelides, R.J., Zebker, H.A., **Zheng, Y.**, (2019). An Algorithm for Estimating and Correcting Decorrelation Phase from InSAR Data Using Closure Phase Triplets. *IEEE Transactions on Geoscience and Remote Sensing*, <http://doi.org/10.1109/TGRS.2019.2934362>
- [2] **Zheng, Y.** and Zebker, H.A., (2017). Phase Correction of Single-Look Complex Radar Images for User-Friendly Efficient Interferogram Formation. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 10(6), <http://doi.org/10.1109/JSTARS.2017.2697861>
- [1] **Zheng, Y.** and Zhou, S., (2014). The spatiotemporal variation of the b-value and its tectonic implications in North China. *Earthquake Science*, <https://doi.org/10.1007/s11589-014-0086-8>

Refereed conference publications:

- [2] **Zheng, Y.**, Zebker, H.A., and Michaelides, R.J., (2020) "A Physics-Based Decorrelation Phase Covariance Model for Effective Decorrelation Noise Reduction in Interferogram Stacks," *2020 IEEE International Geoscience and Remote Sensing Symposium*, <http://doi.org/10.1109/IGARSS39084.2020.9323237>
- [1] Zebker, H.A. and **Zheng, Y.** (2016), Robust and efficient InSAR deformation time series processing, *2016 IEEE International Geoscience and Remote Sensing Symposium*, <http://doi.org/10.1109/IGARSS.2016.7729827>

Works in progress:

*Graduate student mentee

- [2] Mishra, E.*, Jian X.*, and Zheng, Y., in preparation, Monitoring slow slip events in Central Cascadia using InSAR
- [1] Jian, X.*, Mishra E.*, and Zheng, Y in preparation, Quantifying groundwater storage changes in California's San Gabriel Basin from 1992 to 2026 using multi-mission InSAR

Invited talks/presentations at professional meetings, seminars

- 2026 John D. Pigott Colloquium Series, University of Oklahoma, Norman, OK, October 8 (expected)
- 2026 SEG/AGU Hydrogeophysics workshop, Boise, Idaho, July 20-22
- 2026 COMET+ Webinar Series, University of Leeds, UK, remote, February 26.
- 2025 Department of Sustainable Earth System Sciences, University of Texas, Dallas, September 4.
- 2024 NASA Solid Earth Team "Core 2.0" workshop, Washington, DC, September 16-18
- 2024 SMU Energy, Environment and Natural Resources Colloquium, Dallas, TX, May 2.

- 2023 Geoclub Seminar, University of Texas, Dallas, November 10
- 2023 Roy Huffington Department of Earth Sciences, South Methodist University, October 6.
- 2022 Department of Geosciences, University of Texas, Dallas, November 3.
- 2022 NISAR Community Science Workshop, Pasadena, CA, August 17.
- 2022 School of Earth Sciences, Ohio State University, April 13.
- 2022 Department of Earth Sciences, University of Hong Kong, April 3.
- 2022 Department of Earth and Atmospheric Sciences, Indiana University, January 26.
- 2021 Kimbell School of Geosciences, Midwestern State University, December 6.
- 2021 SCEC Community Geodetic Model Workshop, remote, November 29.
- 2020 Seismological Laboratory, California Institute of Technology, November 20.
- 2018 Annual AGU Fall meeting, San Francisco, CA, Dec 15-19

Presentations at professional meetings (first author only)

- Aug 2026 Zheng, Y., Simons, M., Integrating long-term InSAR monitoring into local groundwater management: Insights from the Greater Los Angeles, California. IEEE IGARSS
- Dec 2025 Zheng, Y., Fattahi, H., Modeling, Prediction, and Retrieval of Surface Soil Moisture from InSAR Closure Phase. AGU Fall Meeting
- Dec 2025 Zheng, Y., Simons, M., Integrating long-term InSAR monitoring into local groundwater management: Insights from the Hollywood Basin, California. AGU Fall meeting
- Aug 2025 Zheng, Y., Fattahi, H., Modeling, Prediction, and Retrieval of Surface Soil Moisture from InSAR Closure Phase. IEEE IGARSS
- Dec 2023 Zheng, Y., Simons, M., Ups and Downs of Beverly Hills, California – Analysis of Surface Deformation Derived from InSAR: 1992-2023. AGU Fall Meeting
- Sep 2023 Zheng, Y., Fattahi, H., Modeling soil moisture with cumulated closure phase of interferometric SAR measurements. Fringe 2023.
- Mar 2023 Zheng, Y., Simons, M. Ups and Downs of Beverly Hills, California: Analysis of surface deformation from InSAR: 2015 – 2023. GAGE/SAGE 2023 Community Science Workshop
- Dec 2022 Zheng, Y., Fattahi, H., Modeling soil moisture with cumulated closure phase of interferometric SAR measurements. AGU Fall Meeting.
- Dec 2021 Zheng, Y., Fattahi, H., Agram, P., Simons, M., Rosen, P., On Closure phase and Systematic Bias in Multi-looked SAR Interferometry. AGU Fall Meeting.
- Dec 2021 Zheng, Y., Simons, M., Investigating land surface displacements over the San Gabriel Valley, California. AGU Fall Meeting.
- Jun 2021 Zheng, Y., Fattahi, H., Agram, P., Simons, M., On Closure phase and Systematic Bias in Multi-looked SAR Interferometry. Fringe Workshop.
- Dec 2020 Zheng, Y., Fattahi, H., Agram, P., Simons, M., Assessing closure phase and its impact on InSAR time-series. AGU Fall Meeting.
- Sep 2020 Zheng, Y., Zebker, H.A. and Michaelides, R.J., A Physics-Based Decorrelation Phase Covariance Model for Effective Decorrelation Noise Reduction in Interferogram Stacks. IEEE International Geoscience and Remote Sensing Symposium.
- Dec 2019 Zheng, Y. and Zebker, H.A., Are redundant interferograms really redundant? On the use of redundant interferograms to reduce noise. AGU Fall Meeting.
- Dec 2018 Zheng, Y. and Zebker, H.A., Slow Slip Events in Cascadia: Observation from Sentinel-1. AGU Fall Meeting.
- Dec 2017 Zheng, Y., and Zebker, H.A., Retrieving Ground Deformation Associated with Cascadia Slow Slip Events Using Sentinel-1 Data. AGU Fall Meeting.
- Dec 2016 Zheng, Y. and Zebker, H.A., Crustal deformation associated with Cascadia slow slip events from InSAR time-series, AGU Fall Meeting
- Jul 2014 Zheng, Y. and Zhou, S., The spatiotemporal variation of the b-value and its tectonic implications in North China, International Workshop on Statistical Seismology

External funding for original investigations

Grants awarded

Summary: Total funding awarded as PI: \$858,045; My share: \$471,655

1. Precise Localization and Measurement of Slow Slip Events in Central Cascadia with InSAR, **Yujie Zheng (PI)**, NSF-CRESCENT, 2025-2026, \$29,952, 100% PI share (\$29,952)
2. Four-Dimensional Characterization of Groundwater Recharge from Atmospheric Rivers: A combined Geodetic and Seismic Analysis in Greater Los Angeles, **Yujie Zheng (PI)**, Shujuan Mao (Co-I), NASA, 2025-2028, \$828,093, 53% PI share (\$441,703)

Grants pending

1. Multi-Frequency NISAR Closure Phase and Radar Observables for Monitoring Managed Water Stress in Water-Limited Agriculture, **Yujie Zheng**, NASA-NISAR, 2027-2029, \$658,123, selectable pending funding availability, 100% PI share
2. Tracking Hydrologic Change at the Brine-Freshwater Transition Zone of Salar de Atacama Using NISAR and Complementary Earth Observations. **Yujie Zheng**, NASA-ECIP, 2027-2029, \$299,517, under review, 100% PI share
3. Constraining Slow Slip in Cascadia with InSAR. **Yujie Zheng**, Eshanta Mishra*, NASA-FINNEST, 2027-2029, \$149,025, under review, 100% PI share

Teaching and Mentoring

Postdoctoral supervision:

1. Rishabh Dutta 2025 - present

Doctoral advisement/direction:

- | | | |
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| 1. Eshanta Mishra | 2024 - present | InSAR based Monitoring of Slow Slip Events in the Cascadia Subduction Zone |
| 2. Xiang Jian | 2024 - present | Characterizing Groundwater Recharge Mechanism in Greater Los Angeles Using Multi-Mission InSAR |
| 3. Mauritz Feldbrugge | 2026 - present | Pending |

Classroom teaching:

2024	Spring	GEOS 5v08.004*	Imaging Radar: Principles and Geophysical Applications
2024	Fall	GEOS 4395*	Remote Sensing Geophysics
2025	Fall	GEOS 4395	Remote Sensing Geophysics
2025	Fall	GEOS 5328	Imaging Radar: Principles and Geophysical Applications
2026	Fall	GEOS 4395	Remote Sensing Geophysics

*New courses

Service

University/School/Department Services

- Member, SESS Hydrology Search Committee, 2025-2026
- Member, SESS scholarship committee, 2026

- Undergraduate Advisor, 2024-present
- Member, SESS Seminar Committee, 2023-2024
- Member, NSM strategy committee, 2023-2024
- Member, NSM teaching workload committee, 2023

Professional Service

- Journal reviewer
Nature Communications; Remote Sensing of Environment; Water Resources Research; IEEE Transactions on Geoscience and Remote Sensing; IEEE Transactions on Parallel and Distributed Systems; Journal of Geodesy; IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing; Computers & Geosciences; Earth and Space Science; IEEE Geoscience and Remote Sensing Letters; Remote Sensing Applications: Society and Environment; Remote Sensing in Earth System Science.
- Grant reviewer
NASA Experimental Program to Stimulate Competitive Research, ad hoc reviewer, 2017
NASA FINESST Program, panelist, June 4th to June 6th, 2024
- Conferences
Session Chair, Methodological Aspects in InSAR/DInSAR Analysis , IGARSS (2025)
AGU Geodesy Section OSPA Committee (2024)
Primary Convenor and Chair, Recent Advances in SAR/InSAR, AGU (2024 – present)